

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 4th Year 1st Term Examination, 2018.
URP 4175
(Resettlement Planning)

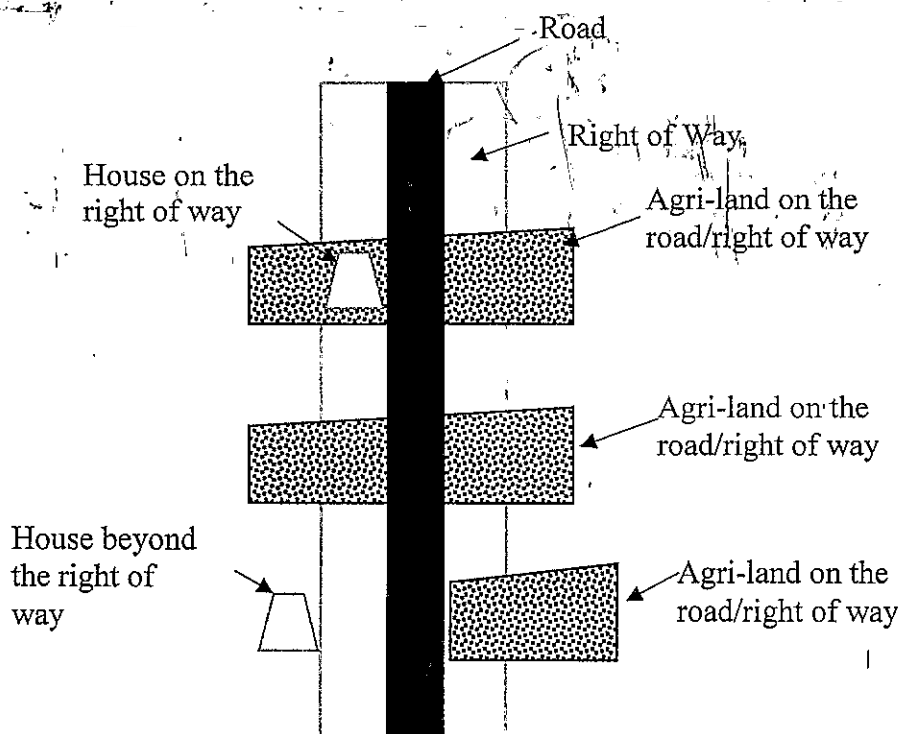
Full Marks: 210

Time: 3.00hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.
iii) Instructions for the candidate: Figure in 2(d) is not to scale

Section – A

1. (a) What are the key components of Resettlement Plan? Why is Resettlement Planning a major concern all over the world? (08)
(b) "The distinction between voluntary and involuntary resettlement often is blurred"- explain. (07)
(c) How can environmental disaster (e.g., draught) become a major cause of conflict and population displacement? (08)
(d) Provide a summary of major government challenges concerning population displacement and resettlement in Bangladesh. (12)
2. (a) What are the major risks of poor resettlement planning for the government and people? In reference to these risks explain the guiding principles of a good resettlement planning? (10)
(b) What is a Resettlement Action Framework? What are the major components of a Resettlement Action Framework? (08)
(c) Write short notes on the following resettlements: (i) Land-based livelihoods, (ii) Wage-based livelihoods, (iii) Enterprise-based livelihoods. (09)
(d) Suggest appropriate Resettlement/compensation for the following figure: (08)



3. (a) What is the difference between land acquisition and requisition? How does the constitution of Bangladesh guarantees resettlement for land acquisition? (12)
(b) Acquisition and Requisition of Immovable Property Ordinance (ARIPO), 1982 is the only instrument for resettlement practice in Bangladesh. Mention the major limitations of the ARIPO. (07)
(c) Padma Multi-purpose Bridge is a land mark project that is displacing a significant number of people by acquiring land. Mention the major concerns for the resettlement practice and future consequence. (11)
(d) Why do people usually refuse to accept compensation? (05)

4. (a) Why Jamuna Bridge is a milestone for resettlement planning in Bangladesh? (10)
Write a short summary on WB Operational policy.
- (b) What were the major limitations of the Jamuna Bridge resettlement planning? What were the major learning to take with from this infrastructure project for future infrastructure projects? (15)
- (c) Draw a chronological account of the indigenous people's conflict with the Bangladesh government in the hilly regions. (10)

Section – B

5. (a) Provide recommendation to address the loopholes for finalizing land price in 1982 ARIPO Ordinance. (10)
- (b) "The present resettlement policies are adequate."- Give your arguments for or against this statement. (10)
- (c) What are the matters that are needed and not needed to be considered in determining compensation? (15)
6. (a) Describe the various Resettlement Planning Instruments and Financial Assistance provided to the borrower. (15)
- (b) "The eligibility criteria for benefits is itself a major short coming"- Explain with the help of a suitable example. (10)
- (c) Describe the followings: (10)
(i) No-worse-off criterion, (ii) Gratitude money.
7. (a) Describe Grievance Redress Mechanism. (10)
- (b) What are the steps to undertake for GRM? (10)
- (c) With the help of an example, show the difficulties and lessons learned in any large resettlement project. (10)
- (d) Explain why beneficiaries may not take GRM seriously. (05)
8. (a) Compose a framework integrating the prospectus of the IRR (Impoverishment Risk and Reconstruction) model and the principles of Resettlement Economics taking into consideration the limitations of Acquisition and Requisition of Immovable Property Ordinance, 1982. (15)
- (b) Write a case study showing the lessons learned and difficulties during an Involuntary Resettlement. (15)
- (c) What are the advantages of integrating IRR model with current resettlement practices? (05)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning

BURP 4th Year 1st Term Examination, 2018

URP 4161

(Project Planning and Management)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable value for any missing data.

Section – A

1. (a) Distinguish between ordinal annuity and annuity due. 08
(b) Find the operating cash flow from the following table. 10

	Year 1 - 5
Project revenues (growth rate, 0%)	240,000
Cost of goods sold (60% of revenues)	144,000
Cash operating expense	5,000
Depreciation	40,000
Taxes (30%)	15,300

- (c) What do you mean by cost of capital? 07
(d) Delta limited has an obligation to redeem Tk. 500 million worth of bonds 6 years hence. How much should the company deposit annually in a sinking fund account where in it earns 14% interest to cumulate Tk. 500 millions in 6 years time? 10
-
2. (a) "A taka today is more valuable than a taka a year hence"- justify the statement. 05
(b) What do you mean by break-even point? 05
(c) The Batpur Group Limited had revenues of Tk. 150 million in 2000, which increased to Tk. 1500 million in 2010. What was the compound growth rate in revenue? 10
(d) Suppose, Khulna Shipyard road has to be reconstructed. Now prepare a log frame as part of Development Project Proposal (DPP) for reconstruction of Khulna Shipyard road. 15
3. (a) Why does the discounted payback period get priority over payback period? 07
(b) Consider a firm which employs equity and debt in equal proportions and whose cost of equity and debt are 14% and 6% respectively. What is the cost of capital? If the firm invests Tk. 100 million which earns a rate of return of 12%, compute the ROI and ROE. Do the equity shareholders benefit? 10
(c) What do you mean by sensitivity analysis? Discuss the advantages and disadvantages of sensitivity analysis. 10
(d) Differentiate between simple and compound interest with necessary example. 08

4. (a) Mention different sources and measures of risk associated with a project. 13
 (b) What is meant by cost of capital? Consider the following debenture of Multiplex limited. 10

Face value: Tk. 1000

Coupon rate: 12%

Remaining period of maturity: 4 years

Current market price: Tk. 1040

Calculate the approximate yield to maturity of this debenture.

- (c) Cash flow forecast for John's Flour Mill project: 12

	BDT in thousands	
	Year 0	Year 1 - 10
1. Investment	20,000	
2. Sales		18,000
3. Variable costs (66.67% of sales)		12,000
4. Fixed costs		1,000
5. Depreciations		2,000
6. Pre-tax profit		3,000
7. Taxes (33.33% of pre-tax profit)		1,000
8. Profit after tax		2,000
9. Cash flow from operation		4,000
10. Net cash flow		4,000

If the cost of capital is 12% calculate:

- I. Accounting break-even sales
- II. Financial break-even sales

Section – B

5. (a) Elaborate the following concepts with necessary examples. 10
 (i) Vision (ii) Mission (iii) Policy (iv) Plan
- (b) What do you mean by Project Management Cycle? Illustrate in brief. 10
- (c) Discuss the various plans that are prepared in planning phase of Project Management Cycle. 15
6. (a) What do you mean by Result-Based Management/Monitoring (RBM)? 08
- (b) Discuss various types of monitoring with necessary example. 12
- (c) Discuss various types of evaluation with necessary example. 15
7. (a) What is a Log frame? 08
- (b) Discuss in brief the steps of a Logical Framework Approach based project design. 15
- (c) Prepare a Log frame for an urban sanitation project. 12
8. (a) Discuss the steps of stakeholder analysis of a project. 10
- (b) Illustrate the structure and functions of Planning Commission in Bangladesh. 12
- (c) Discuss the steps and process of conflict management in a development project. 13

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 4th Year 1st Term Regular Examination, 2018.
URP 4171
(Legal Basis of Planning)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Figures in the right margin indicate full marks.
iii) Answer any **three** questions from each section in separate script.

Section – A

1. a. Suppose you are working in an urban development project (linear park development in city) as a planner. Mention a name of legal tool that is a must follow for this project. Describe the importance of this legal tool. (10)
b. Describe three hierarchical system of urban planning in Bangladesh for post liberation period. (15)
c. What are the shortcomings of Water Act, 2013? (10)
2. a. In which situation, the legal city/town authority should impose "Betterment fee". Provide an example aligned with your opinion. (10)
b. "Town Planning" is an important function of Paurashava Ordinance, 1977. Describe its functions briefly according to the ordinance. (10)
c. Provide two real field examples on violation of planning standards in Bangladesh using two different legal tools. (15)
3. a. Show the setback codes with necessary diagram. (08)
b. Mention the necessary steps of building construction approval process according to Building Construction Act, 1996. (20)
c. Which types of shafts can be used for high rise building construction according to BNBC? Give explanations with necessary diagrams. (07)
4. Write short notes (any five): (35)
 - a) FAR and Building Coverage Ratio
 - b) Major features of BNBC
 - c) Building requirements in flood prone area
 - d) Smoke draft barrier
 - e) Head Room Clearance
 - f) Fiscal Zoning
 - g) BNBC standards for access road and walkways

Section – B

5. a. Write down the importance of studying legal aspects of planning for the students of Urban and Regional Planning. (15)
b. Define the following terms with appropriate examples: (20)
i. Law, ii. Ordinance, (iii) Regulations, (iv) Act, (v) Rules
6. a. Discuss the Statutes & Enforcement Authorities involved in Land Administration and Management in the process of Acquisition of land. (10)
b. Make a list of maps and plans necessary to attach with the application for the approval of private housing project. (10)
c. Critically examine and analyze different rules for Building Construction in different countries. (15)

7. a. Minor Acts are framed on a particular situation or problem. Generally (10)
Minor Acts are related with the development activities, public
administration and public finance; establishment of these Acts follows a
systematic way. Now, state and discuss the formulation process of Minor
Act.
- b. Discuss the suggestions for raising objections against penalties involved (10)
with land acquisition procedures.
- c. Write down the problems of Minor Act. (05)
- d. Which factors are responsible for lengthy process of land acquisition and (10)
compensation? Suggest some measures to solve this problem.
8. Write short notes on the followings (any five): (35)
- a) Eminent domain
 - b) Difference between acquisition and requisition
 - c) Process of acquisition
 - d) Obsolete regulations
 - e) Rules for building construction in Japan
 - f) Private Housing Project discussion consultancy procedure
 - g) Planning regulations of Bangladesh
-

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 4th Year 1st Term Regular Examination, 2017
URP 4111
(Utility Services and Infrastructure Planning)

Full Marks: 210

Time: 3.0 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Define infrastructure. Write down the demographic factors that can influence the infrastructure decision? (14)
- i) Classify the concept of Goods in Economics based on Excludability and Rivalry in a tabular format. (05)
- ii) "Investment in infrastructure can boost Economic Growth and Reduce Poverty" -- Explain the statement with necessary examples. (08)
- iii) How do demand side constraints make it difficult to recover the costs of new infrastructure investment? (08)
2. (a) What are the sources of financing for infrastructure operations and maintenance? Explain different ways of imposing user fees. (10)
- (b) Define the following terms associated with infrastructure: (15)
 - (i) Spillover benefits
 - (ii) Negative externalities
 - (iii) Earmarked Account
 - (iv) Time Inconsistency
 - (v) PPP
- (c) Explain the concept of BDO, BOO, BOT and concession with examples (10)
3. (a) What is Pareto improvement and how is the concept of Pareto improvement related with the economic evaluation of infrastructure project? (13)
- (b) Infrastructure investments can crowd out and crowd in private investment-How? (05)
- (c) Why is it important to set up guidelines for infrastructure service delivery? Write down the differences between output and input based guidelines. (10)
- (d) What are the Seven dimensions of measuring infrastructure demand? Explain in short each of them. (07)
4. (a) Mention the benefits of appropriate maintenance of Infrastructure service delivery with at least two realistic examples (10)
- (b) Khulna Development Authority (KDA) has received a development fund from GoB to construct road infrastructures around the city. Based on potential number of users, construction and operation cost there are four available alternatives to KDA for construction. The alternatives are: Ring road, Flyover, Bypass road and City Corridor. Pair wise comparison of the mentioned criteria and the Ranking Alternatives are given in the following tables: (25)

Criteria	Number of Users	Construction cost	Maintenance Cost
Number of Users	1/1	1/2	3
Construction cost	2/1	1/1	4/1
Maintenance Cost	1/3	1/4	1/1

Number of Users	Ring road	Flyover	Bypass road	City Corridor
Ring road	1/1	1/4	4/1	1/6
Flyover	4/1	1/1	4/1	1/4
Bypass road	1/4	1/4	1/1	1/5
City Corridor	6/1	4/1	5/1	1/1

Maintenance Cost	Ring road	Flyover	Bypass road	City Corridor
Ring road	1/1	2/1	5/1	1/1
Flyover	1/2	1/1	3/1	2/1
Bypass road	1/5	1/3	1/1	1/4
City Corridor	1/1	1/2	4/1	1/1

Construction Cost	Cost in BDT (in millions)
Ring road	34
Flyover	24
Bypass road	28
City Corridor	27

Select the best construction alternative for KDA using analytical hierarchy Process (AHP) method.

Section – B

5. (a) What are the objectives of water supply? (03)
 - (b) Enumerate the general considerations for planning and designing of a water supply system. (10)
 - (c) "Water Demand Management depends on some essential tools and techniques" – Explain them with examples. (10)
 - (d) Estimate average daily water demand for a town having 110,000 population. Assume industrial use 10%, institutional & commercial use 15%, public use 5% and livestock 10% of domestic demand. Take per capita consumption of 50 l/day and leakage to be 5%. (12)
6. (a) Discuss with examples the different approaches for efficient Water Demand Management in residential areas. (10)
 - (b) Write down the advantages and disadvantages of "Gravitational" water transmission and distribution system. (05)
 - (c) What are the driving forces of rain water harvesting in urban areas? (10)
 - (d) Design a service reservoir for a town if Q_{daymax} is 4000 m³. Two pumps are working at a constant rate of 50 litres/second. Determine for how many hours pumping should be done. Daily fire demand is 5% of daily total water demand. (10)
7. (a) Describe different principles and policies undertaken in Bangladesh to achieve sustainable water and sanitation system. (15)
 - (b) Mention different community managed water supply systems in Bangladesh. (10)
 - (c) What are the major challenges prevailing in water and sanitation sectors in Bangladesh? (10)
8. (a) Define sewage, sewer and sewerage. (03)
 - (b) Write down the advantages and disadvantages of Combined Sewerage systems. (10)
 - (c) Mention different design methods of storm sewer. Explain one of them. (10)
 - (d) A 100 hec drainage basin containing 70 hec net residential area with average 5 dwelling units per hec with 5 residents, and 30 hec zoned commercial area. Determine the design flow for a sewer servicing this area. Take wastewater generation for Residential = 250 l/capita/day, Commercial = 1800 l/hect/day, Peak l & l allowance = 10000 l/hect/day. (12)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 4th Year 1st Term Regular Examination, 2016

URP 4141
(Environmental Planning and Management)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any THREE questions from each section in separate scripts.
ii) Figures in the right margin indicate full marks.

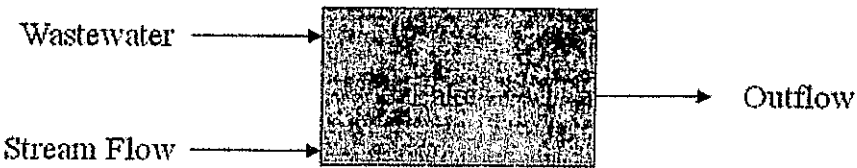
Section – A

1. (a) Discuss the nexus among population growth, poverty and environmental degradation with necessary example. (15)
(b) How do the Technology and Policy can mediate the above nexus to ensure environmental sustainability? Illustrate with necessary example. (12)
(c) Draw your concept on Environmental Kuznet Curve with necessary example and its limitation. (08)
2. (a) Discuss the advantages and disadvantages of various command and control / regulatory instruments of environmental management. (10)
(b) Describe various Market Base Instrument (MBIs) of environmental management with necessary example. (15)
(c) Draw your concept on persuasive instrument of environmental management. Discuss various ways how a persuasive instrument works to ensure environmental sustainability. (10)
3. (a) Illustrate the concept Tragedy of the Commons with necessary example. (08)
(b) Discuss the steps of environmental risk assessment and management with necessary example. (12)
(c) Mention various method of valuation of environment goods. Discuss in brief the Contingent Valuation Method (CVM) of valuation of environmental goods with necessary example. (15)
4. (a) Discuss one important convention and one important protocol that aimed at environmental protection. (05)
(b) What do you mean by internationalization of environmental externalities? Illustrate with necessary example. (10)
(c) Draw your concept on 5 "Rs" that help reducing over consumption. (10)
(d) Draw your concept on various values of environmental quality/goods. (10)

Section – B

5. (a) Differentiate between EIA and SEA. (05)
(b) What are the influential factors that act behind the effectiveness of EIA process? Write down the ways to describe the impacts in EIA process. (10)
(c) When is it required to conduct full EIA? Write down the basic steps of full EIA study. (10)
(d) Explain in brief the different types of Impacts and their attributes that are considered in EIA process. (10)
6. (a) Why a planner should know about Environmental Planning and Management? Explain in brief. (10)
(b) "Cities contribute considerably to environmental problems in local as well as global context." Justify the statement. (10)
(c) What do you mean by environmental management plan? Discuss in brief the steps of preparing an environmental management plan. (15)

7. (a) Why should reversibility, unit of measurement, and probability be considered in impact prediction? (05)
- (b) Mention the uncertainties of impact predictions. (05)
- (c) What is sensitivity analysis? Illustrate with appropriate example. (10)
- (d) A completely mixed lake receives two inflows: natural stream flow $0.1 \text{ m}^3/\text{s}$, wastewater discharge $0.024 \text{ m}^3/\text{s}$ and has a constant volume of $2 \times 10^6 \text{ m}^3$. Provided that the wastewater has $10 \text{ mg/L NH}_3\text{-N}$ and stream has $1 \text{ mg/L NH}_3\text{-N}$. (15)



Bacteria in the lake convert NH_3 to NO_3^- by a process called nitrification. $-r_N = k \cdot C_N$, where k = a first-order rate constant = 0.03 day^{-1} and C_N = concentration of ammonia-nitrogen mg/L . Find the lake and outflow $\text{NH}_3\text{-N}$. Ammonia is very toxic to fish, $1 \text{ mg/L NH}_3\text{-N}$. Does the amount of natural nitrification in the lake allow wastewater discharge of $23 \text{ mg/L ammonia-N}$?

8. (a) Write down the advantages and disadvantages of checklist methods of impact identification. (05)
- (b) Significance of an impact = [Environmental Value x Impact Magnitude]. Explain the equation to calculate the significance of an impact on a biophysical system or component of the environment while conducting EIA. (15)
- (c) In an EIA process, primary, secondary and subsequent impacts (initial condition, consequent condition and their effects) are identified. Total impacts are classified into three categories viz. Branch 1, 2 and 3. Actions and impacts of three Branches along with their magnitude and importance with probability of occurrence are also shown in the following. (15)
- Using Network methods of impact identification, Calculate the network index for each branch along with grand index and provide your insights on the results.

Possible Adverse Impacts		
Initial Condition	Consequent Condition	Effects
Increased surface runoff (E)	Flooding (H)	Gullyng and Erosion (K)
Pollution of Ground Water (F)	Degradation of Water Supply (I)	Health Hazard (L)
Removal of Topsoil (G)	Decreased Fertility (J)	Death of Flora (M)

Relation Between Action and Impacts of three Branches				
Action	Impacts			Branches
A →	(F) →	(I) →	(L)	1
B →	(E) →	(H) →	(K)	2
C →				
D →	(G) →	(J) →	(M)	3

Relation among different type of impacts, their magnitude and importance with probability of occurrence				
Impacts	Impact Score (1-10 Interval Scale)		Probability of Occurrence	
	Magnitude	Importance		
E	5	3	B → E(0.8)	D → E(0.7)
F	2	5	A → F(0.5)	
G	3	4	C → G(0.3)	D → G(0.4)
H	4	5	E → H(0.7)	
I	2	9	F → I(0.6)	
J	2	5	G → J(0.8)	
K	3	7	H → K(0.7)	
L	2	10	I → L(0.9)	
M	1	6	J → M(0.8)	